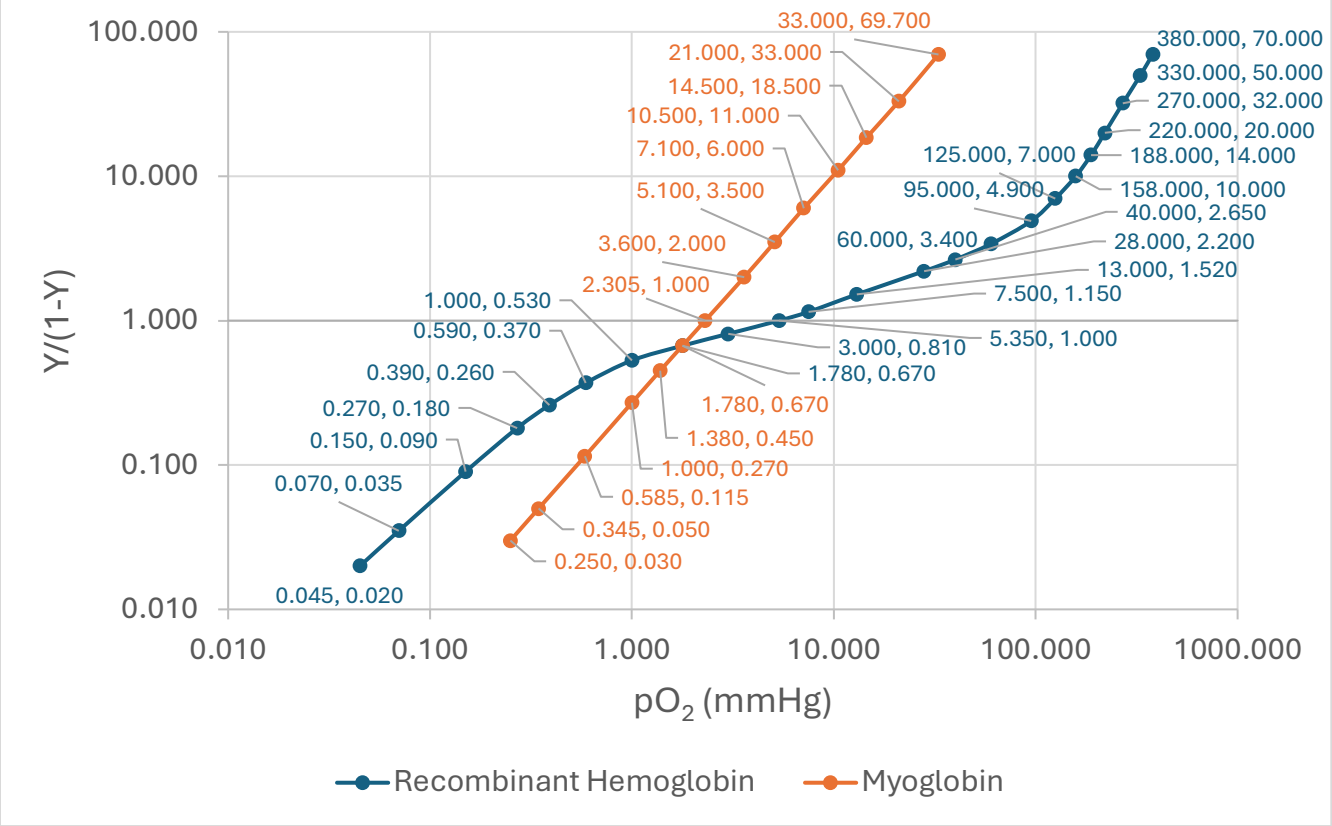


Image description

A Hill plot of oxygen binding to myoglobin and a form of recombinant hemoglobin. Y is the fractional oxygen saturation, and pO_2 is the partial pressure of oxygen in millimeters of Hg. A baseline is shown at the vertical axis value of 1. Each data point is labeled with its exact x and y coordinates.

Task description

An oxygen equilibrium experiment of a mutant hemoglobin gave a distinct biphasic oxygen equilibrium curve as shown in the image, reproduced by superimposing the curves of two independent subunits with different oxygen affinities. An oxygen equilibrium curve was also obtained for myoglobin. The temperature was set at 25 degrees Celsius. Calculate the partial pressure of oxygen at half saturation (P_{50}) for hemoglobin and myoglobin.

Partial pressure at half saturation (P_{50}) means the partial pressure of oxygen (pO_2) at which half of the available ligand-binding sites are occupied. P_{50} values are not shown on the graph. To solve the problem, obtain coordinates from the graph, plug them into an equation, and calculate the P_{50} values.

Data points are explicitly labeled, and gridlines are shown to allow you to unequivocally determine the correct and necessary data points for your calculation. The x-coordinate is pO_2, and the y-coordinate is Y/(1-Y). Select one coordinate pair from the hemoglobin curve and one coordinate pair from the myoglobin curve to proceed. You do not need the Hill coefficient, that is, the slope of the Hill plot, to solve this problem.

Output answer in the following format: Hemoglobin: <value rounded to 3 decimal places> mmHg; Myoglobin: <value rounded to 3 decimal places> mmHg

Final answer

Hemoglobin: 1.887 mmHg; Myoglobin: 3.704 mmHg

Step-by-step solution

1. As a first step, it may help to transform the axes of the graph to see where the hemoglobin line and myoglobin line intersect with the y axis (check sheet 3 of the data file). These intersections depict the $\log P_{50}$ values to be plugged into an equation later. Grab their coordinates. Alternatively, on the graph shown above, locate the data points that have pO_2 value of 1.000 mmHg on the hemoglobin curve and the myoglobin curve. These points are: (1.000, 0.530) for Hb, and (1.000, 0.270) for Mb.

2. Take the log of each coordinate value:
   For both points: $log\ (pO_2)\ =\ log\ (1.000)\ =\ 0.000$
   Point on hemoglobin: $\log\left(\frac{Y}{1-Y}\right)_{Hb} = \log\ (0.530)$
   Point on myoglobin: $\log\left(\frac{Y}{1-Y}\right)_{Mb} = \log\ (0.270)$
   Keep these values in log form until the final calculation to not introduce errors due to early rounding of the decimal places.

   Thus, the logarithmic coordinates of the mentioned points are: (0.000, log (0.530)) on the hemoglobin curve, and (0.000, log (0.270)) on the myoglobin curve. Let A be the point (0.000, log (0.530)), and B be the point (0.000, log (0.270)).

3. The correct Hill equation for this problem is (references 1 - 3):

$$y = \frac{(pO_2)^n}{(P_{50} + (pO_2)^n)}$$

   Linearize this equation by dividing y by $1 - y$, canceling out values, and taking the log of both sides. The result is:

$$\log\left(\frac{Y}{1-Y}\right) = n * log(pO_2) - \log P_{50}$$

   where:
   Y is the fractional oxygen saturation
   n is the Hill coefficient
   pO_2 is the partial pressure of oxygen (mmHg)
   P_{50} is the partial pressure of oxygen at half saturation (mmHg)

4. The x-coordinates at both points A and B are: $log(pO_2) = 0.000$
   The equation in the previous step becomes:
   $\log\left(\frac{Y}{1-Y}\right) = n * 0.000 - \log P_{50}$
   $\Longrightarrow \log\left(\frac{Y}{1-Y}\right) = -\log P_{50}$
   This equation describes the y-coordinate when the x-coordinate is 0.
5. Using y-coordinate of point A:
   $\log\left(\frac{Y}{1-Y}\right) = -\log P_{50} = \log(0.530)$

   $\Longrightarrow \log P_{50} = -\log(0.530)$
   $\Longrightarrow P_{50} = 10^{-\log(0.530)}$
   $\Longrightarrow P_{50} = 1.887$ (mmHg)
   This is the P_{50} for hemoglobin.
6. Using y-coordinate of point B:
   $\log\left(\frac{Y}{1-Y}\right) = -\log P_{50} = \log(0.270)$

   $\Longrightarrow \log P_{50} = -\log(0.270)$
   $\Longrightarrow P_{50} = 10^{-\log(0.270)}$
   $\Longrightarrow P_{50} = 3.704$ (mmHg)
   This is the P_{50} for myoglobin.
7. Final answer:

   Hemoglobin: 1.887 mmHg; Myoglobin: 3.704 mmHg

References

(1) Hill AV. The possible effects of the aggregation of the molecules of haemoglobin on its dissociation curves. J Physiol. 1910; 40:4–7.

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(3) Goutelle S, Maurin M, Rougier F, et al. The Hill equation: a review of its capabilities in pharmacological modelling. *Fundam Clin Pharmacol*. 2008;22(6):633-648. doi:10.1111/j.1472-8206.2008.00633.x.